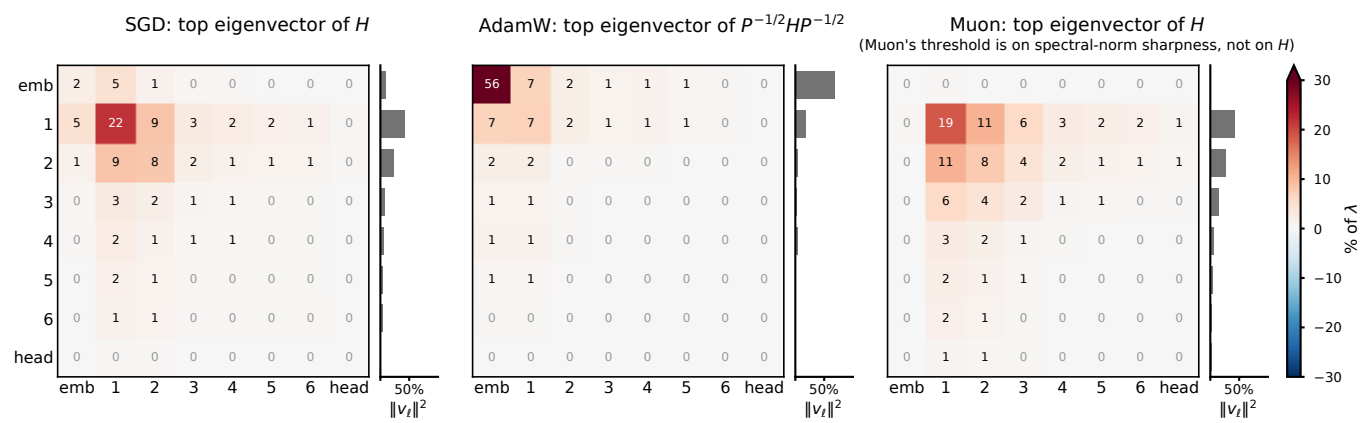
SGD: top eigenvector of $H$
AdamW: top eigenvector of $P^{-1/2}HP^{-1/2}$
Muon: top eigenvector of $H$
(Muon's threshold is on spectral-norm sharpness, not on $H$)
emb
1
2
3
4
5
6
head
50%
$\|v_\ell\|^2$
% of $\lambda$
30
20
10
0
−10
−20
−30